

STEPCUE · REQUIREMENTS

StepCue requirements

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	STC-REQ-001	Version	0.5
Status	Draft	Date	2026-10-01
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stepcue		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-01	Amish Chadha	Status from STC-CAL-001 v0.3 after the design for construction (STC-DDR-003, draft); R10 and R14 figures updated, no status changed
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R6 audio-route target 0.3 s; R9 met with 2.5 mm EVA base
0.3	2026-09-25	Amish Chadha	Targets adopted by Amish 2026-09-25 (STC-DDR-001); R7 audio route and R14 unit restated; TRL 3 status from STC-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

StepCue requirements

Amish adopted targets R1 to R15 on 2026-09-25 (STC-DDR-001, decision D8). R7 and R14 are restated to match decisions D1 (one instrumented insole) and D5 (haptic cue by default, audio through a phone or earbuds). R6 is restated with a separate target for the audio route, decided by Amish on 2026-09-25 (STC-DDR-002, item O2). The status column gives the TRL 3 position from the calculation note STC-CAL-001 v0.3, after the design for construction changes of STC-DDR-003 (draft, open for Amish's review); nothing here has been measured.

Table 1. Requirements and TRL 3 status.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (STC-CAL-001)
R1	Sense plantar loading at the main contact regions	5 or more pressure sites (heel, lateral midfoot, first and fifth metatarsal heads, hallux), each sampled at 100 Hz or more	Design review; sampling calculation	Met: 5 FSRs at 104 Hz. Four sites exceed the 20 N sensor range at peak load, so force above 20 N is lost; timing and center-of-pressure features remain
R2	Sense foot motion for the freeze band	6-axis IMU at 100 Hz or more, covering the 3 to 8 Hz freeze band and the 0.5 to 3 Hz locomotor band	Datasheet; sampling calculation	Met: module IMU at 104 Hz, 0.50 Hz bins
R3	Detect freezes quickly	Median delay from freeze onset to detection 2 s or less	Offline replay of labeled data	At risk. Method floor 1.50 s with one window, 2.50 s with the confirmation R5 needs (synthetic)
R4	Detect freezes reliably	Episode sensitivity 80 % or more; window specificity 85 % or more	Offline replay of labeled data	At risk. Published pressure-only results are about 77 to 80 % and 83 to 85 %
R5	Avoid nuisance cues	1 false cue or fewer per 10 min of ordinary walking	Offline replay; later supervised trial	At risk. Needs 5 or more confirming windows at 85 % specificity
R6	Start the cue promptly	Haptic cue starts 0.1 s or less after a detection; audio through a phone or earbuds starts 0.3 s or less after a detection	Firmware timing calculation	Met: haptic 41 ms; audio route about 0.18 to 0.28 s (estimate, 20 ms margin at worst)
R7	Deliver a rhythmic cue the wearer can feel or hear	Haptic pulses at the wearer's baseline cadence (adjustable 60 to 130 per minute); where the wearer chooses audio, beeps of 60 dB(A) or more at the ear through a paired phone or earbuds	Calculation from datasheets; later bench measurement	Met: 203 Hz pulses stay distinct at 130 per minute; earbuds exceed 60 dB(A). Felt intensity through a sock cannot be shown on paper
R8	Stop cueing when walking resumes	Cue stops within 3 regular steps, or after 15 s at most	Firmware logic review	Met by design (3 steps about 1.7 s)
R9	Keep the shoe comfortable	Insole stack 5.0 mm or less; no rigid part thicker than 1 mm under the heel or metatarsal heads	Parametric model; later pressure mapping	Met: 4.50 mm with the 2.5 mm EVA base (4.7 mm at the upper EVA tolerance); FSRs 0.46 mm
R10	Keep the heel pod light and small	Pod 35 g or less; body no larger than 45 x 40 x 20 mm	Parametric model; later weighing	Met: 27.2 g; 42 x 37 x 15 mm (20.3 mm deep including the clip)
R11	Run a full day and more	2 days or more between charges at 16 h per day of wear	Power budget calculation	Met: 12.5 days nominal, 7.4 days conservative
R12	Usable with reduced dexterity	Pod clips on and off one-handed; USB-C charging with the pod off the shoe; one large button to pause cues	Design review; later user session	Not verifiable at TRL 3. Clip removal about 6 N by calculation; 10 mm button
R13	Keep data with the wearer	Detection and cueing on the device; optional event log exported over USB or BLE by the owner only; no cloud	Design review	Met by design; the phone is used only for optional audio

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (STC-CAL-001)
R14	Low cost and buildable	Parts cost \$200 or less per unit, a unit being one instrumented insole and one heel pod; no custom rigid PCB for the first build	Priced BOM	Met: \$89.94, \$110.06 under the \$200 value-engineering target
R15	Safe to wear	Protected lithium cell outside the shoe; no exposed conductors; skin-contact materials suitable for footwear	Design review	Met by design

■ Requirements not met or at risk

No requirement is not met at TRL 3. R6 (audio route) and R9, listed here in STC-REQ-001 v0.3, are now met after Amish's 2026-09-25 decisions O2 and O3 (STC-DDR-002).

- **R3, R4 and R5, at risk.** The confirmation that R5 needs pushes the detection delay past R3, and published pressure-only accuracy sits at the edge of R4. None can be verified without a labeled plantar-pressure data set.
- **R12, not verifiable at TRL 3.** One-handed use needs a session with users, which is TRL 4 or later and on hold.

■ Assumptions

- Freeze-related leg trembling sits in a 3 to 8 Hz band, compared with 0.5 to 3 Hz for normal stepping; this is the basis of the widely used freeze index ([Sweeney et al., 2019](#); [freeze index reliability, JNER 2023](#)).
- Cueing near the person's own cadence helps, while cueing well below it can make freezing worse ([Sweeney et al., 2019](#), summarizing Arias and Cudeiro 2010 and Lee et al. 2012).
- Shoe size about EU 42 (US men's 8.5) for the massing model; the insole is cut to size in practice.
- Wear of 16 h per day with overnight charging.
- One instrumented insole, worn on the side where the wearer reports freezing most (decision D1).